

CHAPTER 6. MARKUPS FOR EQUIPMENT PRICE DETERMINATION

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CHAPTER 6. MARKUPS FOR EQUIPMENT PRICE DETERMINATION

6.1 INTRODUCTION

This chapter of the technical support document (TSD) presents DOE's method for deriving transformer prices. The objective of the equipment price determination is to estimate the price paid by the customer or purchaser for an installed transformer. Purchase price and installation cost are necessary inputs to the life-cycle cost (LCC) and payback period (PBP) analyses. Chapter 8 presents the LCC calculations; section 8.2.1 describes how the LCC uses purchase price and installation cost as inputs.

Purchase prices for distribution transformers are not generally known. Transformers are specialty items, often custom-built with unlisted prices. The engineering analysis (Chapter 5) provided the manufacturer selling prices for the units included in the LCC analysis. DOE derived a set of prices for each transformer design produced by the engineering analysis by applying markups to the manufacturer selling price in the form of markup equations. These markups represent all the costs associated with bringing a manufactured transformer into service as an installed piece of electrical equipment at a customer's site.

6.2 OVERVIEW OF MARKUP EQUATIONS

Depending on the purchasing environment, DOE used different markup equations to capture the various markups in the supply chain between the manufacturer and the customer. For example, electric utilities (except for the rural electric cooperatives) typically purchase liquid-immersed transformers through manufacturer representatives or distributors. The manufacturer selling price plus the distributor markup is generally the utilities' price for transformers. Dry-type transformers go through several additional marketing or handling steps before they are installed by the end-use purchaser.

Liquid-type distribution transformers have a seven percent markup, accounting for distributor markup.¹ This markup is eliminated for a fraction of cases to account for liquid-immersed transformer sales that are from manufacturers directly to utilities. The fraction of cases is determined by the amount of electricity reportedly sold by IOUs in EIA's Form 861, is 82%.

The manufacturer selling prices for dry-type transformers include two price markups: a distributor markup of 15 percent and a contractor materials markup of 10 percent. DOE based these markups (expressed as average multipliers) on *RS Means Electrical Cost Data Online 2011*.² The distributor markup converts the manufacturer selling price to the distributor price and the price paid by the electrical contractor. This distributor markup covers the costs of the distribution business, including sales labor, warehousing, overhead, and profit. Then the contractor applies a markup to the distributor selling price to cover contractor overhead and profit.

For both liquid-immersed and dry-type transformers, DOE added sales tax, an installation labor and equipment markup, and installation costs. In the previous distribution transformer rulemaking DOE analyzed shipping costs as one of the markups used to determine installed equipment price. In this Preliminary Analysis the markups for shipping costs have been moved into the engineering analysis; this is described in greater detail in chapter 5. Using *RS Means Electrical Cost Data Online 2011*, DOE estimated a contractor markup of 1.10, which is used to convert the distributor selling price to a contractor price. Then the installation cost is added as the cost of labor, equipment, and materials (other than the transformer itself) needed to install a distribution transformer. Finally, by weighting the sales tax for each individual State by its population, DOE calculated a national weighted average sales tax of 6.9 percent.³ DOE developed several empirical equations for estimating installation costs by following these steps mentioned above.

6.3 ESTIMATION OF INSTALLED PRICE

In order to estimate the installed price for distributor transformers, DOE applied the following equation, which describes the steps in the distribution channel of transformers:

$$\text{Installed Price} = M_{tax} \times \{M_{L\&E} \times L\&E + M_{Mat} \times [M_{Dist} \times ManPrice]\}$$

Where:

<i>Installed_Price</i>	=	the final installed price of the transformer (2010\$),
<i>M_{tax}</i>	=	the factor that accounts for sales tax, estimated to be 1.069, ³
<i>M_{L&E}</i>	=	the factor that accounts for the markup on direct installation labor and equipment costs,
<i>L&E</i>	=	the installation, direct labor, and equipment costs (2010\$), adjusted to 2010\$ using the gross domestic product (GDP) price deflator from the <i>U.S. Bureau of Economic Analysis (BEA)</i> . ⁴
<i>M_{Mat}</i>	=	the factor that accounts for the contractor markup on the purchase of the transformer from the distributor,
<i>M_{Dist}</i>	=	the average distributor markup factor, and
<i>ManPrice</i>	=	the manufacturer's selling price (2010\$).

DOE estimated markups on transformers by fitting a linear cost function to the *RS Means* electrical cost data (see section 6.3.3). The *RS Means* data break down the total installed cost for transformers in terms of four cost components:

1. materials: the unit material cost, which includes mounting hardware, but not overhead or profit;

2. labor: labor cost required for installation, including unloading, uncrating, hauling within 200 feet of the loading dock, setting in place, connecting to the distribution network, and testing;
3. equipment: equipment rentals necessary for completion of the installation; and
4. overhead and profit (O&P): installation overhead and profit expenses for the contractor (for dry-type transformers only).

RS Means lists the first three cost components separately and then has an additional column listing the total costs including O&P. As defined by *RS Means*, this figure is the sum of the bare material cost plus 10 percent for profit; the bare labor cost plus total overhead and profit; and the bare equipment cost plus 10 percent for profit.

6.3.1 Estimation of Pole Replacement Costs

In evaluating design options and the impact of potential standard levels, DOE examined the potential for new standards for distribution transformers to lessen the utility or performance of these products. Stakeholders mentioned in their comments to DOE that the more efficient transformers that are heavier could have lessened utility due to impacts on utility pole requirements for overhead transformers. DOE estimated the additional installation costs, based on cost data provided by stakeholders, to mount the single-phase, pole-mounted, liquid-immersed transformers whose designs would require an upgrade to the pole due to increased transformer weight.

DOE included a pole-replacement cost function as a part of the installation cost equation for DL2, which covers pole-mounted transformers. Because not all transformer installations require a change-out of existing equipment even in the most extreme case, DOE assumed a maximum change-out fraction. DOE selected 25 percent as the maximum change-out fraction estimate based on stakeholder input.⁵ This analysis assumed that a pole change-out cost of \$2,000 occurs for up to 25 percent of pole-mounted transformers when the weight of the transformer exceeds 1,000 pounds.¹ DOE also included a transformer pole support cost of \$0.12/lb to account for the cost of a sturdier pole. The pole cost equation is included in the LCC spreadsheet tool for DL2 so that stakeholders can conduct sensitivity analyses regarding pole replacement costs.

6.3.2 Impact of Increased Transformer Weight on Installation Costs

DOE derived the weight-versus-capacity relationship for typical transformers from the design data produced by the engineering analysis. It used the weight-versus-capacity relationship to estimate the transformer weight corresponding to the transformer costs reported in *RS Means*. DOE estimated a scaling relationship between transformer weight and direct installation labor and equipment costs by fitting the correlation between weight and installation costs to a power-law equation.

The method for deriving the weight-versus-capacity relationship uses a typical

transformer weight from the engineering analysis. DOE defined the *typical weight* as the minimum weight plus 20 percent times the weight range, where the weight range is the difference between the minimum and maximum transformer weight for the selected design.

From these data, DOE obtained the following power-law relationship for transformer weight as a function of capacity and basic impulse insulation level (BIL) rating:

$$Weight = 17.31 \times kVA^{0.52} \times BIL^{0.44}$$

Where:

Weight = the weight of the transformer (lbs),
kVA = the capacity of the transformer (kVA), and
BIL = the BIL rating of the transformer (kV).

Although *RS Means* does not provide transformer weights, it does provide transformer capacity and primary voltage. DOE estimated weight from capacity and BIL, which it estimated using primary voltage. DOE then compared the weight to the direct installation costs from the labor and equipment to obtain a power-law relationship.

The following regression performed was the installation direct labor and equipment costs as a function of transformer weight. Data analyzed included all 67 distribution transformer kVA ratings spanning the three *RS Means* electrical equipment categories: “dry type transformer”, “oil-filled transformer”, and “transformer, liquid-filled”. The resulting correlation equation is:

$$L\&E = 38.69 \times Weight^{0.53}$$

Where:

L&E = the installation, direct labor, and equipment costs (2010\$),
Weight = the weight of the transformer (lbs).

6.3.3 Estimation of Markups

DOE performed a regression to disaggregate the overhead and profit associated with installation labor and equipment rental from the overhead and profit associated with the transformer (material) cost. The regression equation is:

$$Total\ Costs\ Including\ O\&P = \alpha + \beta \times Mat + \gamma \times L\&E$$

Where:

Total Costs Including O&P = the sum of all bare costs plus overhead and profit expense (2010\$),
Mat = the material cost (transformer and hardware) (2010\$), and
L&E = the direct labor and equipment costs of installation (2010\$).

After running the regression above, DOE found that the estimated coefficient for the constant term is not significantly different from zero; therefore DOE reran the regression without the constant term. The resulting equation is:

$$Total\ Costs\ Including\ O\&P = 1.10 \times Mat + 1.47 \times L\&E$$

The interpretation of the coefficient of material costs is that when material costs increase \$1, then the total costs including O&P should be expected to increase \$1.10 while holding the other variable constant. Likewise, a \$1 increase in the direct labor and equipment costs will lead to a \$1.47 increase in the total costs including O&P while holding the other variable constant. These two figures were used to allocate overhead and profit expenses to a markup on the distribution price and a separate markup on the direct labor and equipment costs for the installation.

6.3.4 Dry-Type Transformer Installed Price Equation

For dry-type transformers, the result of these analytical steps is a total installed cost equation as a function of the manufacturer selling price, and direct labor and equipment costs, using those markups estimated in section 6.3.3:

$$Installed\ Price = M_{tax} \times \{M_{L\&E} \times L\&E + M_{Mat} \times [M_{Dist} \times ManPrice]\}$$

Where:

Installed_Price = the final installed price of the transformer (2010\$),
M_{tax} = the factor that accounts for sales tax, estimated as 1.069,
M_{L&E} = the factor that accounts for the markup on direct installation labor and equipment costs, estimated as 1.47,
L&E = the installation, direct labor, and equipment costs (2010\$), adjusted to 2010\$ using the GDP price deflator from *BEA*,
M_{Mat} = the factor that accounts for the contractor markup on the purchase of the transformer from the distributor, estimated as 1.10,
M_{Dist} = the average distributor markup factor, estimated as 1.15¹, and
ManPrice = the manufacturer's selling price (2010\$).

DOE applied the installed cost equation by using the manufacturer price and weight from the engineering analysis. For example, according to the engineering analysis, a DL8 (low-

voltage, three-phase, 750 kVA) transformer model with \$1,698.55 of L&E costs listed in *RS Means* has a minimum manufacturer price of \$4,235.47. For this particular transformer, DOE estimated the lower bound of the installed cost to be \$8,396.72, where \$2,669.15 is the installation cost and \$5,727.57 is the sum of the transformer retail price, sales tax, and markups.

6.3.5 Liquid-Immersed Transformer Installed Price Equation

The installed price calculation for liquid-immersed transformers differs from that for dry-type transformers in that the distributor markup used in the equation is 1.07 instead of 1.15 and DOE removed the contractor markup from the equation based on the previous rulemaking.¹ DOE added a new distribution channel to represent the direct sale of transformers to utilities, which account for approximately 81 percent of liquid-immersed transformer shipments. The fraction of utilities that purchase directly from manufacturers is based on the percent of electricity sales by independently owned utilities in the *EIA's Form 861*¹ database. This sales channel removes a distributor markup. The inclusion of this channel reduces the overall markup for liquid-immersed transformers.

$$\text{Installed Price} = M_{tax} \times \{M_{L\&E} \times L\&E + [M_{Dist} \times \text{ManPrice}]\}$$

Where:

<i>Installed_Price</i>	=	the final installed price of the transformer (2011\$),
<i>M_{tax}</i>	=	the factor that accounts for sales tax, estimated as 1.069,
<i>M_{L&E}</i>	=	the factor that accounts for the markup on direct installation labor and equipment costs, estimated as 1.47,
<i>L&E</i>	=	the installation, direct labor, and equipment costs (2010\$), adjusted to 2011\$ using the GDP price deflator from <i>BEA</i> ,
<i>M_{Dist}</i>	=	the average distributor markup factor, estimated as 1.07, and
<i>ManPrice</i>	=	the manufacturer's selling price (2011\$).

As with the dry-type transformers, DOE applied the installed cost equation by using the manufacturer price and weight from the engineering analysis. For example, according to the engineering analysis, a DL4 (medium-voltage, three-phase, 225 kVA) transformer model with \$2,309.64 of L&E costs listed in *RS Means* has a minimum manufacturer price of \$5,305.36. For this particular transformer, DOE estimated the lower bound of the installed cost to be \$9,697.87, where \$3,629.44 is the installation cost and \$6,068.43 is the sum of the transformer retail price, sales tax, and markups.

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